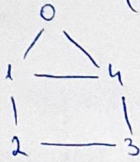


P5 - Hamiltonian Cycle in undirected graph

Hamiltonian cycle $\rightarrow$ visits each vertex once and returns to starting vertex.

Eulerian cycle $\rightarrow$ visits each edge once " - " —

Graphical representation: 1:
($\exists$ H. cycle)



Input file 1: graph1.txt

```
5 6
0 1
0 4
1 2
1 4
2 3
3 4
```

Manual execution 1:

Start:

path: [0, -, -, -, -]

visited = {0}

Step 1: Last vertex: 0

Try vertex: 1

Edge 0-1 exists $\Rightarrow$ add 1 $\Rightarrow$ path: [0, 1, -, -, -]

visited: {0, 1}

Step 2: Last vertex: 1

Try vertex: 2

Edge 1-2 exists $\Rightarrow$ add 2 $\Rightarrow$ path: [0, 1, 2, -, -]

visited: {0, 1, 2}

Step 3: Last vertex: 2

Try vertex: 3

Edge 2-3 exists $\Rightarrow$ add 3 $\Rightarrow$ path: [0, 1, 2, 3, -]

visited: {0, 1, 2, 3}

Step 4: Last vertex: 3

Try vertex: 4

Edge 3-4 exists $\Rightarrow$ add 4 $\Rightarrow$ path: [0, 1, 2, 3, 4]

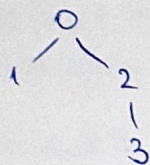
visited: {0, 1, 2, 3, 4}

Final: All vertices are in the path.

Edge 4-0 exists

$\Rightarrow$ Hamiltonian cycle: found: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 0$

Graphical representation 2:
(~~is~~ H. cycle)



Input file 2: graph2.txt:

```

4 3
0 1
0 2
2 3
  
```

Manual execution 2:

Start: path = [0, -, -, -]
visited = {0}

Branch 1: • Last vertex: 0

Try vertex: 1

Edge 0-1 exists $\Rightarrow$ add 1. $\Rightarrow$ path = [0, 1, -, -]
visited = {0, 1}

• Last vertex: 1

Try vertex 2

~~is~~ edge 1-2

• Try vertex 3:

~~is~~ edge 1-3 $\Rightarrow$ backtrack $\Rightarrow$ path: [0, -, -, -]
visited: {0}

Branch 2: • Last vertex: 0

Try vertex: 2

~~is~~ edge 0-2 $\Rightarrow$ add 2 $\Rightarrow$ path: [0, 2, -, -]
visited: {0, 2}

• Last vertex: 2

Try vertex: 3

~~is~~ edge 2-3 $\Rightarrow$ add 3 $\Rightarrow$ path: [0, 2, 3, -]
visited: {0, 2, 3}

• Last vertex: 3

Try vertex: 1

~~is~~ edge 3-1 $\Rightarrow$ backtrack $\Rightarrow$ path: [0, 2, -, -]
visited: {0, 2}

• Last vertex: 2

Try vertex: 1

~~is~~ edge 2-1 $\Rightarrow$ backtrack $\Rightarrow$ path: [0, -, -, -]
visited: {0}

$\Rightarrow$ ~~is~~ Hamiltonian cycle.

Vertex 1 has degree 1 and vertex 3 has degree 1

$\hookrightarrow$ In (~~an~~) Hamiltonian cycle, every vertex must have degree at least 2 (≥ 2) inside the cycle.